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[Data Science / ISII / M1]

PROJECT REPORT

A Comparative Study of RAG Techniques for Image Captioning

Data Exploration and Pre-processing

Assigned technique:	Metadata-Filtered RAG
Group:	1
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Abstract

This report documents the first five phases of a data-preparation pipeline for Metadata-Filtered Retrieval-Augmented Generation (RAG) applied to image captioning on MS COCO 2017. The work covers dataset exploration, data cleaning, statistical analysis, image and caption preprocessing, metadata construction, and preliminary text-embedding generation.

After duplicate removal, the local training subset contains 9,999 unique images and the validation subset contains 100 images, both drawn from COCO train2017. A separate test subset contains 1,000 images from COCO val2017. The training and validation subsets contain 50,018 and 503 cleaned captions, respectively. The test subset retains 5,002 original reference captions for evaluation and has a separate cleaned table containing 5,001 captions. The training and validation metadata comprise 10,099 image-level records and 72,998 object annotations. Preliminary text representations were encoded into 384-dimensional, L2-normalised vectors using `all-MiniLM-L6-v2`.

The report distinguishes completed preprocessing checks from remaining tasks. In particular, the clean training subset is one image below the 10,000-image target; the explicit ID-manifest overlap audit and test-set EXIF-orientation verification remain outstanding; the exact five-caption selection rule used for embedding documents must be clarified; and a separate image-embedding pipeline and vector database/index have not yet been demonstrated. Retrieval, caption generation, and final evaluation are outside the scope of the work reported here.

Keywords: MS COCO, image captioning, Retrieval-Augmented Generation, Metadata-Filtered RAG, data preprocessing, metadata, text embeddings.

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General Introduction

0.1 Context

Image captioning is the task of producing a natural-language description of an image. It requires a model to identify relevant visual content and express it as a coherent sentence. Although modern vision-language models can generate fluent captions, they may omit visible objects, introduce unsupported details, or describe attributes and relationships imprecisely. Retrieval-Augmented Generation (RAG) offers a way to supplement generation with information retrieved from an external knowledge source .

0.2 Problem Statement

This project investigates how RAG techniques affect the quality, relevance, and visual grounding of generated captions on a subset of MS COCO. The work documented in this report addresses the preparation question: how can the image, caption, and object-annotation data be cleaned, organised, and represented so that it can support a reproducible Metadata-Filtered RAG pipeline without using evaluation references as retrieval evidence?

0.3 Objectives

The objectives of the reported phases are to:

- inspect the COCO image and annotation structure;
- identify invalid and duplicate images and prepare clean manifests;
- analyse caption characteristics, object frequencies, scene complexity, and object relationships;
- preprocess images and captions while preserving original evaluation references;
- construct one metadata record per image;
- generate preliminary text embeddings and verify their numerical integrity.

0.4 Methodology and Scope

The work follows five consecutive technical phases: data exploration, data cleaning, statistical analysis, data splitting and preprocessing with metadata construction, and embedding preparation. The report describes only operations and outputs documented in the available

project material. The implementation of a vector database, visual embeddings, metadata-filtered retrieval, caption generation, quantitative caption evaluation, ablation experiments, and statistical comparison of generated captions remains future work.

Data Exploration

1.1 Introduction

Data exploration is the first phase of the project. Before any cleaning, embedding or retrieval, the raw data is inspected to understand what it contains and how it is organized. In this phase, the MS COCO 2017 annotation files (captions and object instances) and the local image folders were examined, in preparation for the image captioning and Metadata-Filtered RAG approach. This phase produces no transformed data; it produces knowledge about the data that guides every later decision.

1.2 Objectives of the Phase

The main objectives are to understand the structure of the COCO annotation files, examine the human-written reference captions and how they are linked to images, explore the object annotations and the 80 object categories, compare the original COCO dataset with the local subsets, verify that the local folders match the project specification, check that the test subset images exist and match the val2017 annotations, and identify what the next phases must handle.

1.3 Libraries and Tools

1.4 Dataset Description and Exploration

1.4.1 Annotation files

The project uses two main COCO training annotation files : `captions_train2017.json` (image information and human-written descriptions) and `instances_train2017.json` (object categories and bounding-box annotations). Both follow the COCO format. The caption file has four top-level keys (info, licenses, images, annotations); the instance file has the same keys plus categories. Each image has a unique integer id, and captions and objects are linked to images through the `image_id` field, not through the file name. All later joins therefore use this identifier. The test images come from the official validation split, whose two files (`captions_val2017.json` and `instances_val2017.json`) have the same structure.

Table 1.1: Libraries and tools used in Phase 1

Library / Tool	Purpose	Used for
json	Reading and parsing the COCO annotation files	Loading the caption and instance annotation files (train2017 and val2017)
pathlib	File and directory management	Verifying folders, listing and counting image files, building paths
collections	Grouping annotations by image	Grouping captions by image_id (defaultdict)
pandas	Tabular data manipulation	Counting captions per image, building summary tables
matplotlib, Pillow	Visualisation and image handling	Imported in this phase, used in the following phases
Jupyter Notebook	Interactive development environment	Running and documenting each exploration step

1.4.2 Global statistics

Table 1.2: Global statistics of the original COCO annotations

Component	train2017	val2017
Images	118,287	5,000
Caption annotations	591,753	25,014
Object annotations	860,001	36,781
Object categories	80	80

1.4.3 Captions per image

Every image in both original COCO splits has at least five captions, and almost all have exactly five. However, some images have six or seven captions. The project specification describes five reference captions per image, so any operation that selects five captions for a retrieval document must use a deterministic, documented rule. The cleaned totals reported in Chapter 2 are retained as observed; they must not be described as exactly five captions per image.

1.4.4 Object annotations and categories

The instance file holds 860,001 object annotations over 80 categories, grouped into supercategories such as person, vehicle and outdoor. Each annotation links an object to an image through `image_id` and stores its `category_id`, a bounding box [x, y, width, height] and an

Table 1.3: Number of captions per image

Statistic	train2017	val2017
Mean	5.0027	5.0028
Standard deviation	0.0523	0.0565
Minimum / maximum	5 / 7	5 / 7
Images with exactly 5 captions	117,972	4,987
Images with 6 captions	312	12
Images with 7 captions	3	1

area.

Table 1.4: First ten object categories

ID	Category	Supercategory
1	person	person
2	bicycle	vehicle
3	car	vehicle
4	motorcycle	vehicle
5	airplane	vehicle
6	bus	vehicle
7	train	vehicle
8	truck	vehicle
9	boat	vehicle
10	traffic light	outdoor

The 80 categories cover people, vehicles, animals, food, furniture and household objects. This structured information describes image content independently of text, and it will provide the visual metadata used for filtering and boosting in the Metadata-Filtered RAG approach.

1.5 Original Dataset vs. Local Dataset: Comparison, Organisation and Verification

1.5.1 Comparison with the original dataset

The initial training folder contained 10,000 images. After one exact duplicate was excluded, the clean training manifest contained 9,999 unique images, approximately 8.45% of the 118,287 train2017 images. This is one image below the 10,000-image target in the project specification. The validation subset contains 100 images and is also drawn from train2017. The coverage limitation should be considered when interpreting later retrieval results.

Table 1.5: Original train2017 images vs. local subsets

Dataset	Images	Share of train2017
Original train2017	118,287	100%
Initial training folder	10,000	8.45%
Clean training manifest	9,999	8.45%
Local validation subset	100	0.08%

1.5.2 Dataset organisation and verification

Table 1.6: Dataset organisation and folder verification

Dataset	Images found	Purpose
Training	10,000	Knowledge base: main data for development and retrieval preparation
Validation	100	Hyperparameter tuning and intermediate evaluation
Test	1,000	Final evaluation only, never part of the knowledge base

The training and validation images are stored in separate folders. The training and validation subsets are drawn from COCO train2017, while the test subset is drawn from the official val2017 partition. This source-level separation is consistent with the intended split design. Nevertheless, the three local ID manifests must still be compared explicitly and the overlap counts recorded before the report can claim that the manifest-level leakage audit has passed. The initial folders contained 10,000 training images and 100 validation images; after duplicate removal, the clean training manifest contains 9,999 unique images.

1.6 Test Images and Annotation Matching

The 1,000 test images were taken from the COCO val2017 split. The notebook extracted the image IDs from the local file names and matched them with the IDs in the val2017 caption annotations. All 1,000 images matched, and none was missing from the caption annotations. The notebook also found no missing or empty captions among the 5,002 original captions associated with these images. After whitespace normalisation and removal of one duplicate caption within an image, 5,001 captions remained in the cleaned caption table. The original captions were nevertheless preserved separately for final evaluation.

1.7 Conclusion

The exploration confirmed that COCO provides linked natural-language captions and structured object annotations. The local clean subsets contain 9,999 training images and 100

validation images, while the separate test subset contains 1,000 images from val2017. The explicit comparison of all three local ID manifests remains necessary before reporting a completed leakage audit.

Data Cleaning

2.1 Introduction

Data cleaning is the second phase of the project. It checks that the local data is reliable before any statistics, embeddings or retrieval are built on it. It covers the image files (can they be opened, are any repeated?), the links between images and captions (does every image have its references?), and the caption text itself. The phase excludes invalid samples from the clean lists, never deletes the original files, and records every decision.

2.2 Objectives of the Phase

The main objectives are: verify the integrity of the local image files and detect corrupted images; detect exact duplicate images; prepare clean lists of valid and unique images; check that every image has associated human-written captions; normalise the caption text without changing its meaning; generate clean caption files, image manifests and a cleaning report.

2.3 Libraries and Tools

Table 2.1: Libraries added in Phase 2

Library	Purpose
Pillow (verify, UnidentifiedImageError)	Detecting unreadable images

Only the libraries added in this phase are listed. The libraries introduced in Phase 1 (json, pathlib, collections, pandas and Pillow) are reused. Pillow is now used actively: its `verify()` method and its `UnidentifiedImageError` exception detect unreadable images.

2.4 Main Results

2.4.1 Local Image Inventory

The supported image files (.jpg, .jpeg, .png) of the training and validation folders were collected. This gave 10,100 local images: 10,000 training images and 100 validation images. The 1,000 test images are stored separately and are treated in Section 2.4.8.

2.4.2 Image Integrity Verification

Each image was opened and verified with Pillow. A file that raised an error (unidentified image, unreadable file or invalid value) would have been placed in a separate list of corrupted images.

Table 2.2: Image integrity results

Dataset	Images	Valid	Corrupted
Training	10,000	10,000	0
Validation	100	100	0

No corrupted image was found, so no sample was removed for integrity reasons.

2.4.3 Duplicate Detection

The SHA-256 hash of every valid image was computed. Images with identical hashes are exact byte-for-byte duplicates. The hashing covered the training and validation images together, so it also checks for duplicates across the two splits. One duplicate group was found, containing two identical files, both in the training folder:

- `000000438867.jpg` (retained)
- `000000531721.jpg` (excluded from the clean list)

No duplicate exists between the training and validation sets. One copy is kept in the cleaned training dataset, and the original image files are preserved.

2.4.4 Preparation of Clean Image Lists

After identifying the corrupted and duplicate files, lists containing only valid and unique images were prepared. Corrupted files and duplicate copies are excluded from these lists, without deleting the original files. The training and validation datasets remain separate.

The clean training list therefore contains 9,999 images, one fewer than the 10,000 images of the project specification, and the clean validation list contains 100 images. All following steps use these clean lists.

Table 2.3: Clean image lists

Dataset	Valid images	Corrupted excluded	Duplicates excluded	Clean images
Training	10,000	0	1	9,999
Validation	100	0	0	100
Total	10,100	0	1	10,099

2.4.5 Image-Caption Association

The COCO caption file (118,287 annotated images and 591,753 captions) was loaded. Captions were associated with images through the unique `image_id`; each local file name was first used to look up its ID. Every image of the original annotation file has captions.

Table 2.4: Image-caption association

Dataset	Clean images	With captions	Without captions
Training	9,999	9,999	0
Validation	100	100	0

2.4.6 Caption Cleaning

The caption normalisation was deliberately light so that the meaning is preserved: leading and trailing spaces were removed, repeated whitespace was collapsed to a single space, and empty captions would have been dropped. No caption was lowercased or stripped of punctuation.

Table 2.5: Clean caption files

Dataset	Clean images	Clean captions
Training	9,999	50,018
Validation	100	503
Total	10,099	50,521

The caption counts are slightly higher than five per image: 50,018 training captions against $9{,}999 \times 5 = 49{,}995$ (23 more), and 503 validation captions against 500 (3 more). These differences arise because some COCO images have six or seven captions. The current cleaning outputs retain these additional captions. The exact rule used later to construct one embedding document per image must therefore be stated separately; the report must not imply that every cleaned image has exactly five captions.

2.4.7 Manifests and Cleaning Report

A manifest was generated for each dataset. It lists the dataset name, image ID, file name, image dimensions and image path of every clean image. A cleaning report summarises all operations.

Table 2.6: Cleaning report

Dataset	Original	Valid	Corrupted	Duplicates	Clean	With captions	Without captions
Training	10,000	10,000	0	1	9,999	9,999	0
Validation	100	100	0	0	100	100	0

Five files were created and verified: `train_captions.csv`, `validation_captions.csv`, `train_manifest.csv`, `validation_manifest.csv` and `cleaning_report.csv`.

2.4.8 Test Set: Image-Caption Association

The test set was prepared in a separate notebook from the official val2017 annotations, following the same logic.

Association. All 1,000 test images were matched to the val2017 caption annotations using their image IDs. No image was missing from the caption annotations, and no caption was missing or empty. The original caption table contains 5,002 captions for the selected images.

Caption cleaning and references. For the cleaned caption table, whitespace was normalised and duplicate captions for the same image were removed. This reduced the count from 5,002 to 5,001 (one duplicate removed). The cleaned distribution is: 997 images with five captions, one image (ID 211825) with four captions, and two images (IDs 215259 and 328030) with six captions. Importantly, the notebook preserves the original, uncleaned captions separately as the reference captions for final evaluation; it does not replace them with the deduplicated captions.

2.5 Conclusion

The training and validation data cleaning retained 9,999 and 100 images, respectively. The separate testing preparation notebook processed 1,000 val2017 images, retained 5,002 original reference captions, and produced 5,001 cleaned captions after removing one within-image duplicate. Ten test images have no object annotations, but their image records are preserved.

Statistical Data Analysis and Visualisation

3.1 Introduction

Statistical analysis is the third phase of the project. It studies the cleaned data to describe what it contains: how long the captions are, which words and objects dominate, how complex the scenes are, how objects are arranged, and which objects appear together. This phase does not modify any image or caption. It produces statistics that justify later design choices, in particular the visual metadata used for filtering in the Metadata-Filtered RAG approach.

3.2 Objectives of the Phase

The main objectives are:

- Measure caption length, vocabulary size and word frequencies.
- Identify the most frequent object categories (top 50).
- Measure scene complexity as the number of annotated objects per image.
- Analyse the grammatical structure of the captions (part-of-speech distribution).
- Study the diversity of attributes (colours, shapes and sizes) and contexts (indoor/outdoor, day/night).
- Extract approximate spatial relationships between objects from their bounding boxes.
- Identify which object categories frequently appear together.
- Combine the main results into a final statistical summary.

3.3 Libraries and Tools

Only the libraries added in this phase are listed in Table 3.1. The libraries introduced in the previous phases (json, re, pathlib, collections, pandas and Pillow) are reused. matplotlib, which was only imported in Phase 1, is now used to create and save all the charts.

3.4 Main Results

Table 3.1: Libraries added in Phase 3

Library	Purpose	Used for
nltk	Natural language processing	Tokenising (word_tokenize) and tagging parts of speech (pos_tag), with the resources punkt_tab and averaged_perceptron_tagger_eng
itertools	Iterator building blocks	Generating each pair of objects or categories only once (combinations) for the spatial relationships and the co-occurrence analysis
IPython.display	Rich output in notebooks	Displaying formatted tables and results (display, Markdown)

The analysis uses the cleaned datasets of Phase 2. The text statistics (Sections 3.4.1 to 3.4.4) were computed on the 50,018 cleaned training captions. The object statistics (Sections 3.4.5 to 3.4.8) were computed on the 10,099 cleaned training and validation images, using the 72,998 object annotations that belong to them.

3.4.1 Caption Statistics

The number of words of every training caption was computed, and the vocabulary was built after lowercasing and removing punctuation.

Table 3.2: Caption length statistics (training captions, in words)

Statistic	Value
Captions	50,018
Mean	10.44
Standard deviation	2.38
Minimum	6
25% / 50% / 75%	9 / 10 / 11
Maximum	47

3.4.2 Most Frequent Words

The most frequent words are function words: a (82,570), on (18,333), of (17,119), the (16,884) and in (15,821). After removing a list of stop words, the most frequent content words show what COCO captions describe.

Captions are centred on people (man, woman, people), their postures (sitting, standing, holding) and common settings (street, table).

Table 3.3: Ten most frequent content words

Word	Frequency	Word	Frequency
man	6,435	white	3,069
sitting	4,418	woman	2,876
two	4,085	street	2,657
standing	3,642	table	2,437
people	3,529	holding	2,295

3.4.3 Part-of-Speech Distribution

Captions were tokenised and tagged with NLTK. The percentages are computed over the three categories analysed (nouns, verbs and adjectives).

Table 3.4: Part-of-speech distribution of the training captions

Category	Count	Share
Nouns	184,806	64.07%
Verbs	60,166	20.86%
Adjectives	43,484	15.07%

Nouns dominate, which shows that captions are mainly object-oriented. Verbs describe actions and adjectives add details.

3.4.4 Attribute and Context Diversity

A keyword-based count was applied to the training captions. The counts are word occurrences, not verified visual attributes and not numbers of images. White is the most mentioned colour. Captions describe size far more often than geometric shape (square appears 60 times and round 57 times). Indoor and outdoor context terms exist but are rare compared with objects.

3.4.5 Object Categories (Top 50)

The instance annotations were filtered with the image IDs of the clean training and validation manifests. The 10,099 local images contain 72,998 object annotations, and the distribution is highly imbalanced.

The category person alone accounts for 22,532 of the 72,998 annotations, about 31%. The full top-50 list ends with airplane (453 annotations) and is saved as `top_50_object_categories.csv`.

Table 3.5: Ten most frequent object categories

Rank	Category	Annotations
1	person	22,532
2	car	3,706
3	chair	3,398
4	book	2,040
5	bottle	1,964
6	cup	1,612
7	dining table	1,260
8	bowl	1,179
9	traffic light	1,106
10	handbag	1,066

3.4.6 Scene Complexity

Scene complexity was measured as the number of annotated objects per image.

Table 3.6: Objects per image

Statistic	Value
Images	10,099
Mean	7.23
Standard deviation	7.37
Minimum	0
25% / 50% / 75%	2 / 4 / 10
Maximum	78

The mean (7.23) is well above the median (4), so the distribution is right-skewed: many images contain a few objects and a minority contain dozens. Eighty-five images (0.84%) have no object annotation. They all exist in the original COCO image records, so no information is missing, and they are retained with an object count of zero.

3.4.7 Spatial Relationships

Relationships were derived from the bounding boxes. For each pair of objects in an image, the dominant direction between the box centres gives left, right, above or below, and a pair is labelled overlap when the intersection over union exceeds 0.1. Each pair is processed once. In total, 501,906 pair relationships were extracted.

Horizontal relationships dominate. The most frequent pairs involve the same category, such as person-person (34,486 right and 33,056 left pairs) and car-car, followed by person-chair.

Table 3.7: Distribution of spatial relationships

Relationship	Count	Share
right	177,736	35.4%
left	176,493	35.2%
above	68,045	13.6%
below	57,159	11.4%
overlap	22,473	4.5%

These relationships are geometric approximations, not annotated semantic relations, and the left and right labels depend on the order in which the two objects are annotated.

3.4.8 Object Co-occurrence

Co-occurrence was counted per image on unique categories, over the 10,014 images that have at least one annotation. This gives 1,848 distinct category pairs.

Table 3.8: Most frequent category pairs (number of images)

Pair	Images
car + person	733
chair + person	707
handbag + person	538
dining table + person	461
backpack + person	450
chair + dining table	419

Person appears in 5,489 images (about 55% of the annotated images) and is part of most frequent pairs. The matrix of the 15 most frequent categories shows clear scene groups: chairs, tables, cups and bowls (indoor and dining scenes), and cars, trucks and traffic lights (street scenes). Co-occurrence shows that two categories share an image, not that they interact.

3.4.9 Final Statistical Summary

The last step of the phase combines the main statistical results of the dataset exploration and analysis into a single summary table, which is exported as `final_statistical_summary.csv`. It gives a concise overview of the cleaned dataset, the caption statistics, the object annotations, the scene complexity and the object co-occurrence.

The number of object categories was verified directly in the original instance file (80), because the variable that first held it had been overwritten by an earlier step and gave an incorrect value in the first version of the table. The corrected summary was saved again.

Table 3.9: Final statistical summary

Metric	Value
Training images	9,999
Validation images	100
Total clean images	10,099
Training captions	50,018
Validation captions	503
Total captions	50,521
Average captions per training image	5.00
Average caption length (words)	10.44
Training vocabulary size	9,683
Local object annotations	72,998
Object categories in the original COCO dataset	80
Average objects per image	7.23
Images without object annotations	85
Unique object co-occurrence pairs	1,848

3.5 Conclusion

The descriptive results show that the captions are relatively short and object-oriented, while the person category accounts for approximately 31% of the local object annotations. Scene-complexity and co-occurrence statistics provide useful descriptive signals for designing metadata filters. They do not, by themselves, demonstrate that metadata filtering improves caption quality; that claim must be tested in the later retrieval and generation experiments.

Data Splitting, Image Preprocessing and Metadata Preparation

4.1 Introduction

This phase turns the cleaned data into the shared, reproducible artefacts that all later stages use. It has four parts:

- **Split organisation:** the intended knowledge-base, validation, and test subsets are stored in separate ID files; a direct cross-manifest overlap audit remains to be completed.
- **Image preprocessing:** images get a consistent orientation and colour format.
- **Caption preprocessing:** the original captions are preserved and a cleaned version is added.
- **Metadata construction:** one record per image combines file information, captions and object annotations.

The original files are never modified. All outputs are written to separate folders. Phase 6 contributes its test-set preparation to this phase: matching the selected test images to the val2017 annotations, preserving the original reference captions, extracting object annotations, creating RGB and grayscale versions, building the testing metadata table, and performing final verification. Section 4.3.8 summarises the test-set artefacts.

4.2 Objectives

The objectives of this phase are to:

- Organise the training, validation and test splits and prepare the ID files required for a leakage audit.
- Convert the training and validation images to RGB with corrected EXIF orientation, preserving the originals; verify the test-set orientation step separately.
- Keep raw captions unchanged and store a cleaned version under a clearly named field.
- Link captions and object annotations to each image through the image ID.
- Build and verify one metadata record per image.
- Produce a grayscale version of the images for later experiments.
- Prepare the same artefacts for the test set.

4.3 Main Results

4.3.1 Load the Cleaned Dataset

Before preparing the metadata, the cleaned image manifests and caption files generated during the data cleaning stage are loaded. The training and validation manifests are loaded together with their captions, and their structure is verified.

Table 4.1: Cleaned dataset loaded

Dataset	Images	Captions
Training	9,999	50,018
Validation	100	503

Each manifest has the columns `dataset`, `image_id`, `file_name`, `width`, `height` and `image_path`, and each caption file has the columns `image_id`, `file_name` and `caption`.

4.3.2 Dataset splits

Table 4.2: Dataset splits

Split	Source	Images	Captions	Role
Training (knowledge base)	COCO train2017	9,999	50,018	Retrieval base
Validation	COCO train2017	100	503	Hyperparameter tuning
Test	COCO val2017	1,000	5,002	Final evaluation

The training and validation splits are stored as `train_ids.json` and `val_ids.json`. The testing notebook saves the selected IDs separately in `testing_image_ids.json`. The test images come from the official val2017 partition, whereas the local training and validation subsets come from train2017. This separates the sources at the COCO split level; however, the testing notebook does not itself show an explicit comparison of the 1,000 test IDs against the local training and validation ID manifests. That ID-overlap check should be run and its result recorded before claiming that a complete anti-leakage audit has been performed. The training set has 9,999 images instead of 10,000 because one exact duplicate was removed in Phase 2.

4.3.3 Image-level metadata construction

Captions and object annotations were grouped by `image_id`, which is the only key used to join the data. For each image, the object records store the category name, category ID, bounding box and area. Images with no object annotation keep an empty object list instead of being dropped.

- Images with captions: 10,099
- Images with object annotations: 10,014
- Images without object annotations: 85

4.3.4 Image preprocessing

All 10,099 training and validation images were inspected before conversion.

Table 4.3: Colour modes of the original images

Mode	Images	Share
RGB	10,078	99.79%
L (grayscale)	21	0.21%

Each image was opened with Pillow , its EXIF orientation was corrected, it was converted to RGB, and it was saved as JPEG (quality 95) in separate train and validation folders under the same file name. The 21 grayscale images were converted to three-channel RGB so every image matches the input format of the image encoder.

Verification confirmed that 10,099 images were saved, that all are in RGB mode, and that none is missing or invalid. A report (`image_preprocessing_report.csv`) records the original mode, processed mode, processed path and dimensions of every image, and one manifest was saved per split. Resizing and normalisation were not applied here. They depend on the encoder chosen in the embedding step.

4.3.5 Caption preprocessing

Two caption fields are kept side by side:

- `original_caption`: the human-written text, never modified, used for evaluation.
- `cleaned_caption`: the text with line breaks replaced by spaces, repeated whitespace collapsed, and leading and trailing spaces removed.

The cleaning is deliberately conservative: it does not lowercase, remove punctuation or change any word. Quality checks found no missing and no empty caption, in either field, in either split (50,018 training and 503 validation captions). Preprocessed caption files were saved separately for training and validation, and reloaded to confirm the record counts and columns.

4.3.6 Final metadata

Image information, caption annotations, and object annotations were associated using `image_id` as the common identifier. These data do not have a one-to-one relationship at the annotation level: one image may have multiple captions and multiple object annotations. Therefore, the caption and object records were first grouped by `image_id` and then aggregated into image-level fields. The resulting metadata table contains one row per image while preserving the grouped caption and object information.

The construction and verification process can be summarised in three stages:

1. **Group and aggregate:** image identifiers and paths are combined with the grouped original/cleaned captions and grouped object records. The image ID is the key used to associate the records.
2. **Verify:** the exported table is checked for the expected number of image records, unique image IDs, missing fields, and consistency between the grouped caption/object totals and the source annotation tables. Images with no object annotations are retained with an empty object list.
3. **Export:** the verified image-level table is saved as `metadata.csv`.

This design maintains one metadata record per image without incorrectly imposing a one-to-one relationship on the underlying caption and object annotations.

Table 4.4: Fields of the final metadata file (metadata.csv)

Group	Fields
Identification	image_id, file_name, dataset
Image	processed_path, width, height, original_mode, processed_mode
Captions	original_captions, cleaned_captions, caption_count
Objects	objects, object_count, object_categories

Table 4.5: Verification of the final metadata

Check	Result
Records	10,099 (9,999 training, 100 validation)
Unique image IDs / duplicates	10,099 / 0
Missing values in any column	0
Total captions / mean per image	50,521 / 5.0
Total object annotations / mean per image	72,998 / 7.23
Images with / without objects	10,014 / 85
Missing processed image files	0

4.3.7 Grayscale representation

A grayscale copy of every RGB image was produced with Pillow's L mode, in separate train and validation folders. All 10,099 files were saved and verified. A separate report (`image_preprocessing_report_grayscale.csv`) and a separate metadata file (`metadata_grayscale.csv`, 16 columns) add two fields, `grayscale_path` and `grayscale_mode`, without changing `metadata.csv`. Verification passed: no duplicate IDs, no missing paths, no missing files, and 10,099 images in mode L. The dataset therefore offers two interchangeable image representations.

4.3.8 Test set preprocessing and metadata (COCO val2017)

The test set was processed in a separate notebook. The notebook's opening markdown paragraph still says that 100 images were selected, whereas its executed matching step and final verification report 1,000 images; the opening paragraph should be corrected to 1,000 for consistency. The available notebook results document image validation, RGB conversion, and grayscale conversion. However, the notebook does not show an EXIF-orientation correction operation (for example, Pillow's `ImageOps.exif_transpose`). Therefore, this report does not claim that EXIF orientation was corrected for the test images. This step should be added and rerun, or its absence should be accepted and documented consistently with the chosen image-encoder requirements.

- **Reference captions.** The original captions were preserved for all 1,000 test images, giving 5,002 reference captions in total. They were grouped by image ID and stored without using the deduplicated caption table.
- **Caption quality and cleaning.** The notebook found zero missing captions and zero empty captions. Whitespace normalisation and removal of duplicate captions within the same image reduced the cleaned caption count from 5,002 to 5,001. The cleaned distribution is 997 images with five captions, one image (ID 211825) with four captions, and two images (IDs 215259 and 328030) with six captions.
- **Object annotations.** A total of 7,375 object annotations were extracted for 990 images. Ten images have no object annotations; their metadata records are retained with zero objects. All 7,375 bounding boxes passed the notebook's validity checks.
- **Image quality.** All 1,000 source images were valid JPEG files. No corrupted or invalid-format images were found, and MD5-based duplicate detection found zero duplicate groups.
- **RGB conversion.** All 1,000 images were converted to RGB and saved in the separate `testing_images_rgb` folder. The notebook reports zero conversion errors. EXIF-orientation correction is not demonstrated in the notebook and remains a verification/action item.
- **Grayscale conversion.** All 1,000 images were converted to grayscale and saved in `testing_images_grayscale`. The notebook reports zero conversion errors.
- **Metadata.** The exported `metadata_testing.csv` contains 1,000 records and 11 columns: `image_id`, `file_name`, `width`, `height`, `rgb_path`, `grayscale_path`, `reference_captions`, `num_captions`, `object_categories`, `num_objects`, and `bounding_boxes`.
- **Verification.** The final verification passed all seven checks: expected image count, metadata record count, unique image IDs, RGB processing, grayscale processing, metadata file existence, and testing ID file existence. The notebook reports 1,000 original images, 1,000 processed RGB images, 1,000 processed grayscale images, 5,002 original reference captions, and 7,375 object annotations.
- **Saved files.** `metadata_testing.csv` and `testing_image_ids.json`.

The test data is kept in its own folder and files. The notebook does not show an explicit ID-overlap comparison against the knowledge-base and validation manifests, so that audit remains to be documented.

4.3.9 Splitting and Comparison of the Splits

The training set provides the knowledge base, the validation set is used for tuning hyper-parameters, and the test set (drawn from the official COCO val2017) is reserved for final evaluation. The source COCO partitions are distinct, but an explicit comparison of the local ID manifests is still needed to document the complete anti-leakage audit.

Table 4.6: Statistical comparison of the training, validation and test splits

Statistic	Training	Validation	Test
Source	train2017	train2017	val2017
Images	9,999	100	1,000
Captions	50,018	503	5,002
Captions per image (mean)	5.00	5.03	5.002
Image representations	RGB & L	RGB & L	RGB & L

4.4 Conclusion

The preprocessing phase produced separate training/validation metadata for 10,099 images and a separate testing metadata file for 1,000 images. RGB and grayscale outputs were generated for both sets. The training/validation metadata checks and the seven reported test-file/count checks passed. However, the report does not yet establish a complete cross-manifest ID-overlap audit, and test-set EXIF-orientation correction is not demonstrated. Accordingly, the three subsets are maintained separately, but full split-isolation and preprocessing equivalence should not be claimed until those two items are verified.

Embedding Preparation

5.1 Definition

Embedding preparation converts each knowledge-base entry into a numerical vector so that similar entries can be found by vector similarity. In this phase, each image is represented by a single text document (its captions plus its object categories), encoded with a pre-trained sentence encoder, and stored with an index that links every vector back to its image.

5.2 Objectives

The objectives of this phase are to:

- Build one textual description per image from the cleaned captions and object categories.
- Generate a fixed-size embedding for each description with a pre-trained text encoder.
- Save the embeddings and an index that preserves the image-vector association.
- Verify the saved files (shape, alignment, NaN and infinite values, normalisation).
- Record the embedding configuration for reproducibility.

5.3 Main Results

5.3.1 Text representation

Each image is represented by one text document combining caption text and sorted object categories. The current report describes this document as containing five cleaned captions. However, Chapter 2 reports 50,018 training captions and 503 validation captions, which exceed five captions per image. The implementation must therefore document exactly how the five captions are selected for each embedding document (for example, a deterministic ordering and selection rule). Until that rule is confirmed from the notebook/code, the phrase "five cleaned captions" should be treated as an unresolved implementation detail. The intended document format is:

```
Captions: <caption 1> ... <caption 5>
Objects: <category 1>, <category 2>, ...
```

For image 36, for example, the object part is "person, umbrella". The lists stored in `metadata.csv` were converted back to Python lists before joining. The text encoder is

based on the Sentence-BERT family of sentence embedding models . The embedding output contains 10,099 text descriptions, with no missing or empty description reported. The precise caption-selection rule used to build each description must be verified before the representation can be considered fully reproducible.

This choice has two consequences that the report should state:

1. Every vector summarises an image as a whole (all five captions together), not individual captions.
2. The object categories come from the COCO annotations, so they are metadata of the knowledge base, which is what the Metadata-Filtered RAG needs.

5.3.2 Embedding model and configuration

Table 5.1: Embedding configuration

Setting	Value
Model	sentence-transformers/all-MiniLM-L6-v2
Type	Sentence-BERT text encoder
Embedding dimension	384
Input	embedding_text (captions + objects)
Batch size	32 (316 batches)
Normalisation	L2-normalised, unit-length vectors
Output format	NumPy array, .npy
Libraries	sentence-transformers 6.2.0, transformers 5.22.1, torch 2.15.0

Because the vectors have unit length, cosine similarity and inner product give the same ranking.

5.3.3 Saved artefacts

Table 5.2: Files produced

File	Content
`embeddings/image_embeddings.npy`	Array of shape 10,099 × 384 (approximately 15.5 MB when stored as float32).
`embeddings/embedding_index.csv`	Mapping table containing `image_id`, `file_name`, `dataset`, and `embedding_index` for each vector.

Row i of the array corresponds to the row with `embedding_index` $= i$ in the mapping CSV. This file preserves the vector-to-image association, but it is not itself a vector database or a

searchable vector index.

5.3.4 Verification

The saved files were reloaded and checked. A first-embedding check was also displayed (the first ten values of the vector for image 36), and the image for row 0 was shown next to its grayscale version to confirm that the row-to-image mapping is correct.

Table 5.3: Verification of the saved embeddings

Check	Result
Embedding array shape	(10,099, 384)
Index records	10,099
Unique image IDs	10,099
Missing values in the index	0
NaN	0
Infinite values in the embeddings	0
Image IDs aligned with the metadata order	True
All vectors unit-normalised	True

5.3.5 What these vectors can and cannot do

The embeddings are text-based. They can support text-to-text semantic retrieval from a text query, but they do not encode image pixels and cannot independently represent a new query image. In addition, the saved NumPy array and mapping CSV are preliminary embedding artefacts, not a complete vector database. A searchable index, its index type, similarity metric, stored metadata, database size, and indexing time remain to be implemented and reported.

5.3.6 Test set (COCO val2017)

No text embeddings were generated for the 1,000 test images. Their 5,002 original reference captions are retained for evaluation only and must not be used as retrieval inputs or generation context. The test images must instead be represented through the pipeline's visual analysis step (an image encoder, and optionally a detector), so test-side visual embeddings belong to the next phase. Only the knowledge base is embedded here.

5.4 Conclusion

10,099 metadata records were converted into verified 384-dimensional L2-normalised text embeddings with a correctly aligned index, ready for retrieval.

General Conclusion and Future Work

6.1 Work Achieved

The first five phases established a data-preparation foundation for the Metadata-Filtered RAG image-captioning project. The work examined the COCO annotation structure, checked the local image files, detected an exact duplicate in the training folder, cleaned and associated captions, analysed text and object statistics, prepared RGB and grayscale image representations, constructed image-level metadata, and generated preliminary text embeddings.

The resulting clean development data contains 9,999 training images and 100 validation images, with 50,018 and 503 cleaned captions respectively. The separate test subset contains 1,000 COCO val2017 images. Its original 5,002 reference captions remain preserved for evaluation, while the cleaned caption table contains 5,001 entries. The training/validation metadata includes 10,099 image-level records and 72,998 object annotations. The preliminary text embedding array contains 10,099 vectors of 384 dimensions.

6.2 Relevance to the Next RAG Stages

The image-level metadata provides a structured association between each image, its captions, and its COCO object annotations. This structure can support later metadata-based filtering, provided that the retrieval documents and vector index preserve the same image identifiers and metadata fields. The text embeddings provide a first semantic representation of caption-and-object text. They do not replace the image embeddings required by the multimodal pipeline.

6.3 Limitations and Outstanding Verification

Several requirements remain incomplete or require stronger evidence:

- The clean training manifest contains 9,999 unique images, one fewer than the required 10,000. A valid, unique replacement image should be added only if permitted by the shared dataset protocol.
- A direct pairwise comparison of the training, validation, and test ID manifests has not been documented. The overlap counts must be computed and recorded.

- The test preprocessing notebook demonstrates RGB and grayscale conversion but does not demonstrate EXIF-orientation correction.
- The exact deterministic rule for selecting five captions per embedding document is not stated, despite the cleaned caption totals exceeding five per image.
- The current text embedding array combines training and validation records. A training-only knowledge-base embedding collection and a separate validation collection are required.
- The file named `image_embeddings.npy` contains text-derived vectors. It should be renamed to avoid implying that it represents image pixels.
- The mapping CSV is not a vector database. A FAISS, ChromaDB, or LanceDB index, including its type, metric, dimensions, size, and indexing time, remains to be built.

6.4 Future Work

The next phases should complete the outstanding data checks, restore the required training-set size if authorised, and freeze the validated preprocessing release. The project can then generate image embeddings with the selected image encoder, construct the vector database, implement metadata-filtered retrieval, and connect the retrieved evidence to the selected vision-language captioning model and generative component. Finally, the system should be evaluated on the fixed 1,000-image test subset without using its reference captions for retrieval or generation. The planned experiments should include a no-RAG baseline, the assigned Metadata-Filtered RAG pipeline, quantitative caption metrics, hallucination analysis, confidence assessment, ablation experiments, and appropriate paired statistical tests. These are proposed next steps, not completed results of the present report.